

# TB XYZ Panner

## DETAILED ENGLISH USER MANUAL

A spatial panner with stable left/right PanCore, front/back, height, distance, room cues, headphone HRTF mode, and surround remapping.



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Host-visible endpoints documented: 9

# 1. Product purpose

**A spatial panner with stable left/right PanCore, front/back, height, distance, room cues, headphone HRTF mode, and surround remapping.**

Conventional pan controls only left/right gain. TB XYZ Panner keeps that reliable foundation and adds procedural spectral, level, and reflection cues for height, distance, and front/back perspective. Headphones mode adds KEMAR-based binaural rendering, while surround layouts use speaker azimuth remapping.

## What result it creates

- Reliable stereo left/right placement
- Procedural near/far and up/down cues
- Front/back impression without rewriting pan
- Headphone binaural and multichannel positioning options

## Signal flow

**Input -> PanCore -> front/back, height, distance, and room cue layer -> Stereo or Headphones renderer -> surround azimuth remap when applicable -> Output Trim**

# 2. Complete feature guide

## Top View

Horizontal movement controls Pan; vertical movement controls Front/Back. Top View does not change Distance.

## Side View

Horizontal movement controls Distance and vertical movement controls Height.

## PanCore

Pan ranges from hard left through center to hard right. It remains the stable localization foundation and is not directly rewritten by front/back cues.

## Height

Uses tonal and ceiling/floor reflection balance to suggest elevation. It is a psychoacoustic cue rather than a physical vertical speaker position in stereo.

## Distance and Room

Distance combines attenuation, low-pass damping, and early reflections. Room controls the reflection contribution. The direct path is not simply moved by a modulated delay.

## Render Mode

Stereo is intended for speaker-compatible panning. Headphones engages a compact KEMAR HRIR-based HRTF renderer; individual listeners may perceive front/back differently.

## Surround and Output

On supported 5.1/7.1 outputs, non-LFE channels are remapped by speaker azimuth. Output Trim preserves headroom after spatial processing.

## 3. Quick start

1. Choose Stereo for speakers or Headphones for binaural monitoring.
2. Set Pan in Top View or with the Pan control.
3. Set Front/Back vertically in Top View.
4. Set Height and Distance in Side View.
5. Add Room only as needed.
6. Trim Output and verify on the actual delivery format.

## 4. Practical workflows

### Near overhead object

1. Use a centered or slightly offset Pan.
2. Raise Height in Side View.
3. Keep Distance low and use modest Room.

**Expected result: The source stays direct while gaining an upward cue.**

### Far rear ambience

1. Move Front/Back toward rear in Top View.
2. Increase Distance in Side View.
3. Raise Room and reduce Output if the field builds up.

**Expected result: The source feels farther and less direct with a rearward impression.**

## 5. Automation, gain staging, and session use

- Automate only controls that serve a clear musical transition. Fast movement of frequency, time, pitch, mode, oversampling, or algorithm selectors may intentionally create artifacts or require a host-safe transition.
- Match perceived loudness before judging quality. A louder setting will often appear better even when the processing decision is not better.
- Use the plug-in bypass endpoint for comparison and preserve an unprocessed source or earlier project version.
- Factory programs are starting points. Loading a program changes multiple parameters; save a new DAW preset after adapting it to the source.
- The parameter appendix is captured from the installed VST3 host contract. It lists every host-visible endpoint, its displayed range/options, default, automation status, and identifier.

## 6. Limits, cautions, and troubleshooting

- Exact rear and height localization on two speakers is physically limited.
- The HRTF is not personalized, so front/back confusion is possible.
- Always verify headphone work on speakers and vice versa when both are delivery targets.
- No audible change: confirm the plug-in and relevant sub-block are enabled, Mix/Amount is not at a neutral value, and the source contains energy in the processed region.

- Unexpected level: return input, output, send, feedback, and parallel-mix controls to conservative values, then rebuild the setting one block at a time.
- Automation does not match the panel: reload the project, confirm the DAW is addressing the current plug-in version, and check whether a factory program or state recall replaced values.
- Clicks or transitions: avoid sample-instant changes to algorithm, time, pitch, mode, or oversampling selectors unless the sound is intentional.

## 7. Complete host parameter reference

This appendix contains all 9 parameters exposed by the current installed VST3 to a host. Repeated EQ-band endpoints are intentionally listed rather than collapsed. Discrete options are printed in full when the host contract exposes them.

| Parameter                                | Range / options                                                             | Default     | Host status            | Function                                                                                       |
|------------------------------------------|-----------------------------------------------------------------------------|-------------|------------------------|------------------------------------------------------------------------------------------------|
| <b>Bypass</b><br>VST3 ID 1652125811      | Off, On                                                                     | Off         | Automatable;<br>Bypass | Turns the complete plug-in processing path off or on through the host-visible bypass endpoint. |
| <b>Distance</b><br>VST3 ID 288459765     | 0 % to 100 %                                                                | 0 %         | Automatable            | Sets the named spatial coordinate or psychoacoustic positioning cue.                           |
| <b>Front Back</b><br>VST3 ID 127212208   | B 100 % to F 100 %                                                          | F 100 %     | Automatable            | Sets the named spatial coordinate or psychoacoustic positioning cue.                           |
| <b>Height</b><br>VST3 ID 926454055       | -100 % to 100 %                                                             | 0 %         | Automatable            | Sets the named spatial coordinate or psychoacoustic positioning cue.                           |
| <b>Output Trim</b><br>VST3 ID 873633987  | -24.0 dB to 0.0 dB                                                          | 0.0 dB      | Automatable            | Sets or limits the final output level after the plug-in's main processing.                     |
| <b>Pan</b><br>VST3 ID 110749             | -100 % to 100 %                                                             | 0 %         | Automatable            | Sets the named spatial coordinate or psychoacoustic positioning cue.                           |
| <b>Program</b><br>VST3 ID 1886548852     | Front Focus, Wide Drum L, Wide Drum R, Overhead High, Back Shadow, Far Room | Front Focus | Automatable            | Selects a factory program. Loading a program recalls a coordinated set of plug-in parameters.  |
| <b>Render Mode</b><br>VST3 ID 1193882713 | Stereo, Headphones                                                          | Stereo      | Automatable            | Selects the operating algorithm, response shape, or tonal character for the named block.       |
| <b>Room</b><br>VST3 ID 3506395           | 0 % to 100 %                                                                | 35 %        | Automatable            | Shapes spatial density, resonant spread, or diffuse body for the named process.                |

### Documentation basis

Prepared from the current public catalog, current local product brief and implementation notes where available, existing English manuals where available, and a direct scan of the installed VST3 host contract. Internal implementation details that are not user-facing are intentionally excluded; all user-facing features and host-visible controls are documented.